Alexander Debusk

Module 06 Contribution Journal - Regression and Classification

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L06 Contributions – Alex DeBusk

- Reviewed and formatted the provided Python code to make it easier to read and understand.
- Completed the missing code for the linear regression and logistic regression models.
- Trained both models using the appropriate training data and generated predictions using the test data.
- Used Mean Squared Error to evaluate fare predictions and accuracy to evaluate survival predictions.
- Identified and fixed a NameError caused by NumPy not being imported.
- Reviewed the assignment instructions to make sure the code matched the required tasks and avoided unnecessary additions.
- Completed the knowledge check questions about regression, classification, model coefficients, and evaluation metrics.
- Prepared the reflective journal describing the challenges encountered and the concepts learned during the lab.
- Organized the required filenames for the lab notebook, reflective journal, and contribution journal for Group 4.
- Improved my understanding of how machine learning models use data to make predictions and how their performance is evaluated.

Shareefah Adeola

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CONTRIBUTION JOURNAL – Module 06: Regression and Classification

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Student: Shareefah Adeola

Date: 10/13/2026

I worked through the Titanic dataset exercise involving Linear Regression and Logistic Regression. My contribution included preparing the data, defining the input features and target variables, and dividing the dataset into training and testing sets.

I implemented and evaluated the Linear Regression model to predict passenger fares using Age and Pclass. The model produced a Mean Squared Error of approximately 3,364.92. I also implemented and evaluated the Logistic Regression model to predict passenger survival using Age, Pclass, and Sex, achieving approximately 74.83% accuracy.

During the exercise, I corrected code-formatting and import errors, reran the relevant cells, printed the Linear Regression coefficients, and completed the Knowledge Check questions. I also ran the notebook from beginning to end to confirm that it was executed without errors.

This work strengthened my practical understanding of model training, prediction, evaluation, and Python debugging. I documented my work and learning as part of my individual contribution to L06.

Yilin Leng

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Contribution

For Lab 06 in ITAI 1371, I completed the main coding tasks, model testing, written answers, and documentation on my own. I loaded the Titanic dataset and cleaned the data by removing rows with missing values in the `Age` column. I also changed the values in the `Sex` column into numbers, using 0 for male and 1 for female. For Task 1, I used `train_test_split` to divide the data into training and testing sets, then trained a `LinearRegression` model using `Age` and `Pclass` to predict passenger fares. I evaluated the model using the Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). For Task 2, I used `Age`, `Pclass`, and `Sex` to predict whether passengers survived. I trained a `LogisticRegression` model and used `accuracy_score` to check its performance. The model achieved an accuracy of 74.83%. I also answered the knowledge check questions about regression, classification, model coefficients, and evaluation metrics. Finally, I checked the code in Google Colab to make sure it ran correctly and prepared the completed work as a PDF file for submission.